Inter-City Freight Load Matching Marketplace

Requirements Engineering & UML Use-Case Modelling

PES University - Department of CSE | Lab 1

FUNCTIONAL REQUIREMENTS (FR)

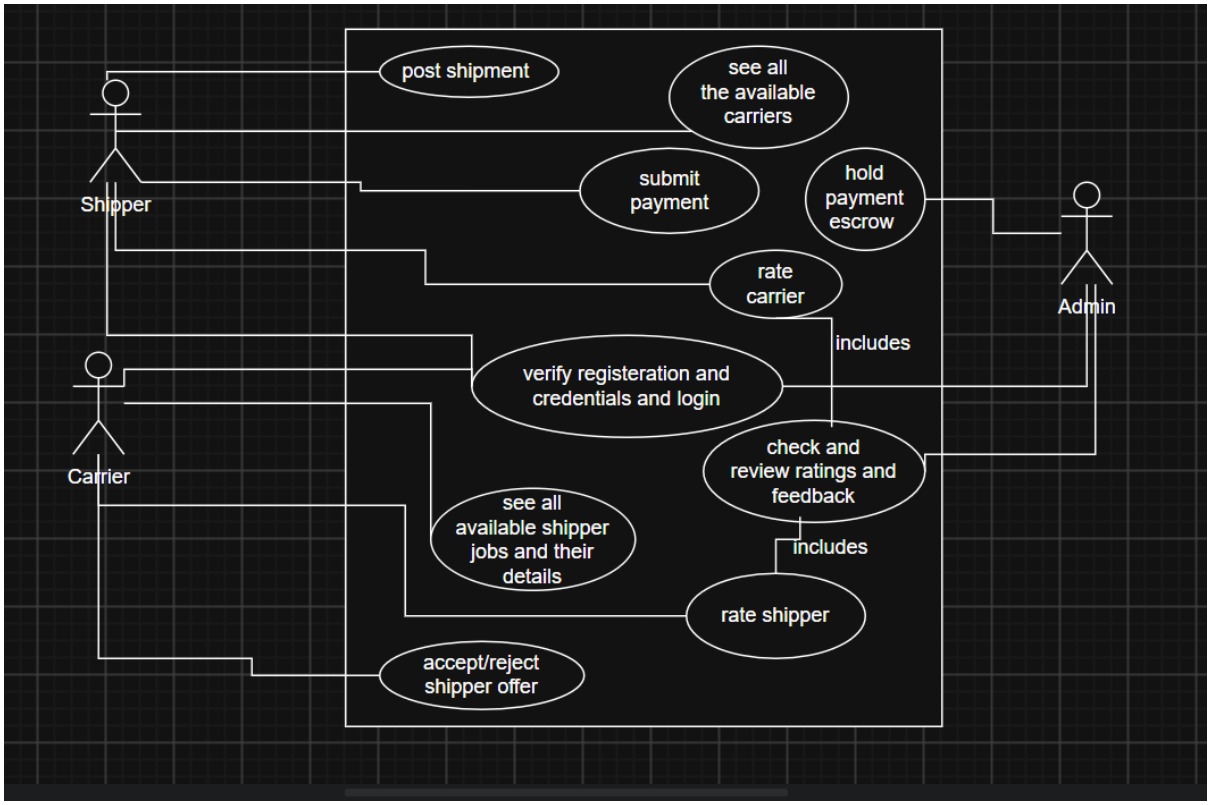
ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall verify and authenticate users (shippers and carriers) through identity validation, permit verification, and business registration before allowing platform access.	High	User registration requires government ID, business license, and tax ID; Unverified users blocked from posting/bidding with clear messaging.	Ensures only legitimate commercial entities participate, reducing fraud risk.
FR-002	The system shall allow shippers to specify shipment details including: source location, destination location, shipment date, cargo type, quantity, special handling requirements (fragile/hazardous), and budget.	High	All fields are mandatory and validated; Posted shipment displays all parameters	Enables precise freight matching and carrier filtering.
FR-003	The system should allow shippers to choose which carrier to choose from all the available ones.	High	Shipper sees carrier bids and can select one; job status updates live to show if carrier accepted or rejected; let shipper only offer the job to only certain available carriers	Enables shippers to assign jobs to preferred carriers and track real-time status of acceptance.
FR-004	The system should show each carrier all the jobs that are offered to it including the respective shippers	High	Carrier views job list with shipper details and can accept/reject each job; shipper is notified instantly of carrier's decision.	Allows carriers to choose profitable jobs and keeps shippers informed of acceptance status.
FR-005	The system shall process payment after a carrier is selected and hold the amount in escrow until proof-of-delivery is confirmed	High	Multiple payment methods accepted (card, bank transfer); Payment confirmed and held; Shipper receives receipt	Protects both shipper and carrier; Builds payment trust.

NON-FUNCTIONAL REQUIREMENTS (NFR)

ID	Type	Description	Priority	Acceptance Criteria
NFR-001	Performance & Scalability	The system shall process and display search results for available freight loads within 30 seconds under normal load and 60 seconds at peak capacity.	High	Load testing confirms <30s response under 1,000 users; <60s at 10,000 concurrent users; Database query optimization documented.
NFR-002	Security & Compliance	All payment transactions and proof-of-delivery data shall be encrypted.	High	Encryption implemented for all sensitive data; Encryption verified through security testing; All data

				transmission uses encrypted protocols.

UML DIAGRAM



Main Use Case:

1. The user registers on the platform as either a **Shipper** or **Carrier**.
2. The user submits the required identity, business, and permit documents for verification.
3. The system validates the documents and grants the verified user access to the platform.
4. The **Shipper** enters and posts the shipment details, such as source, destination, cargo type, quantity, special requirements, and budget.
5. The Shipper views the available carriers and offers the shipment job to selected carriers.
6. The **Carrier** views the offered job along with the shipper and shipment details.
7. The Carrier accepts or rejects the job. If accepted, the Shipper selects that carrier for the shipment.
8. The Shipper makes the payment using an available payment method.
9. The system confirms the payment, holds the amount in **escrow**, and provides a payment receipt.
10. After proof of delivery is confirmed, the payment can be released to the carrier.

Alternate Use Case:

1. The user submits invalid or incomplete verification documents.
2. The system rejects the verification and blocks the user from accessing restricted platform features.
3. The user corrects and resubmits the required information.
4. If the shipment details contain missing or invalid information, the system asks the Shipper to correct them before posting.
5. If a selected Carrier rejects the offered job, the Shipper can select and offer the job to another available Carrier.
6. If no Carrier accepts the job, the shipment remains unassigned until another Carrier is selected.

7. If the payment fails, is cancelled, or contains invalid details, the system informs the Shipper and requests another payment attempt or method.
8. If proof of delivery is not confirmed, the payment remains held in escrow.